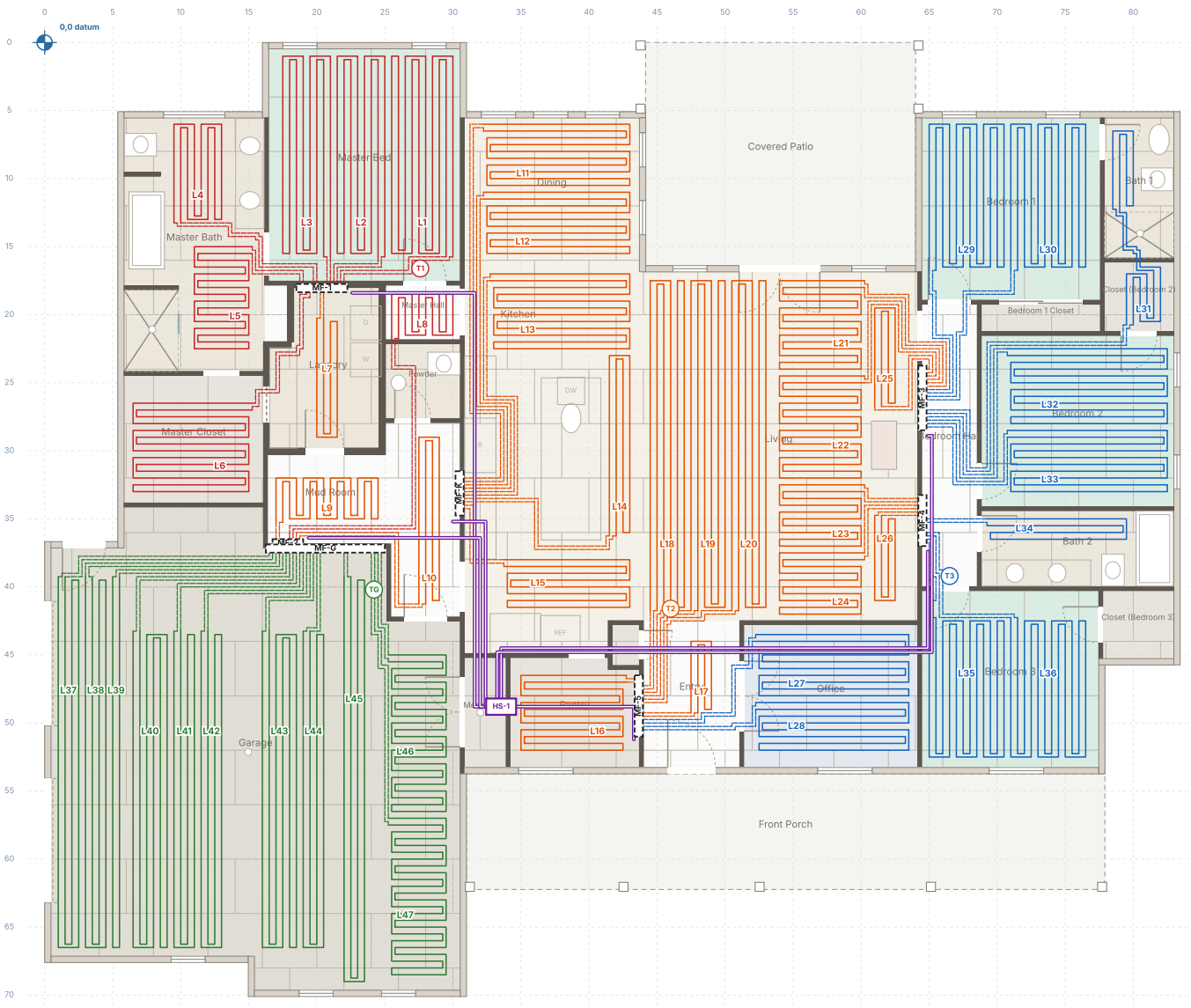
REAR · PATIO



FRONT · PORCH

Zone 1, Master suite (T1) · Zone 2, Great room and utility (T2) · Zone 3, Bedroom wing and office (T3) · Zone G, Garage (held at 50°F) (TG)

----- leaders, supply and return side by side 3" apart · 1" feed pair from HS-1, under the boards · manifold cabinet over its row of 3" ports

4' x 4' nub boards in a running bond from the datum; dots are the 3" puck grid

47 loops, 5,957' of 1/2" PEX, 7 manifolds, no loop over 200'. · Red zone 1, orange zone 2, blue zone 3, green garage. · 52 kBtu/h at design, 110°F supply. · EPS25 Super 4 nub boards on the 3" grid from the datum; nothing crosses.

draft

R-1 Radiant · 64 Stone Sheep Circle

PEX loops on 3" nub boards under the slab: 200' loops, nothing crossing, seven manifolds, four thermostat zones, flows at Powell, WY design (-11°F). App version v19. From the owner's plans (Larson House, 6/11/26).

Notes

1. The layout is drawn on the boards' own grid. EPS25 Super 4 nub boards (4' x 4', 5" thick: 4" of EPS, R-18, under a 1" grid of pucks on 3" centres) go down in a running bond with their puck rows on the 3" lines from the datum at 0,0 (the garage-door wall's outside face and the master bed's rear wall line, as on every sheet), so every tube on this sheet sits in a channel between pucks. Lay the first board's corner at the datum and the rest follow; a board cut at a wall keeps its grid.
2. Tube: 47 loops of 1/2" O2-barrier PEX, none over 200' with its leaders and the 2' rises at the cabinet (L32 is the longest at 190', L17 the shortest at 36'). Fields at 6" o.c. (every other channel), supply and return side by side so each loop's two ends leave its field together; leaders run as 3" pairs through doorways and along walls to their manifold. Nothing crosses anything: the whole layout was checked tube by tube on the grid. One continuous coil per loop, no fittings in the slab; tag both ends with the loop number at the cabinet.
3. Under a wall only where drawn, straight across and only for the length drawn: under the master bed wall into MF-1, under the kitchen wall into MF-K, under the entry wall into MF-5, under the living room wall into MF-4, under the master hall wall into the powder room, and under the bath 1 wall from the bedroom 2 closet. Mark every wall crossing on the boards before the pour so the plate anchors and shots stay 12" off it; everywhere else the leaders go through the doorway.
4. Nothing under the kitchen base cabinets' footprint except the leaders drawn there, nor under the island, fridge, back cabinet, range, washer, dryer, tubs, vanities or the fireplace hearth. 12" clear of the floor drains and shower drains, 3" of a toilet flange; the showers' floors are heated. The bathroom loops heat the open floor; the closets and halls get their heat from the leaders passing through, and the laundry has a small field of its own.
5. Manifolds, seven stations, each with a 3" port position per tube end in one row along its wall (the schedule gives every loop's two port

Open items

- L31 (Closet (Bedroom 2), Bath 1): the floor carries all but about 780 Btu/h of the room at -11°F with 110°F water. Add a small electric towel warmer, or run the boiler hotter on the coldest days.
- Boiler model: HS-1 is a condensing boiler on natural gas or propane (decided 2026-10-13). Pick a combi or a boiler with an indirect tank if it should also make the domestic hot water and replace P-1's 50 gal heater; propane needs the boiler's LP conversion kit.
- Garage: kept at 50°F on its own manifold, pump and thermostat. Decide whether it runs a glycol mix behind a plate heat exchanger, which protects it if the house loses power in a cold snap with the doors open.
- Flooring: the loops assume hard-surface floors. Carpet in a bedroom needs about 10°F warmer water for that loop.
- P-1's hot and cold PEX runs at 6.5" to 8" below the slab top, which is now inside the 5" board layer. Either drop them into the base course under the boards (about 10" down) or cut channels in the EPS for them; P-1 to pick.
- Cabinet depths on S-1: MF-3 and MF-4 recess into the hall's 2×4 west wall (3-1/2" deep; fur locally if the chosen manifold needs more), MF-K into the kitchen wall and MF-5 into the entry wall.
- E-1: an outlet at each of the seven manifold cabinets for the actuators (MF-1, MF-2, MF-G, MF-K, MF-5, MF-3, MF-4), and the boiler, pump and zone controller on circuit 27.
- The EPS25 Super 4 nub boards come 5 to a bundle (80 sq ft), 3 bundles minimum: 55 bundles cover the 4,167 sq ft slab with 5% for cuts. Confirm the lead time; the board layout can start at any corner as long as the grid holds the datum.

numbers, counted from the low end of the row): MF-1 (laundry, on the north wall west of the dryer, 7 loops) serves Master Bed, Master Bath, Master Closet, Laundry; MF-2 (mud room, on the south wall (garage wall), west end, 3 loops) serves Master Hall, Mud Room; MF-G (garage, on the mud room wall (two 7-loop manifolds side by side), 11 loops) serves Garage; MF-K (mud room, east wall (kitchen wall), south of the powder room, 5 loops) serves Dining, Kitchen; MF-5 (pantry, on the east wall (entry wall), 7 loops) serves Pantry, Living, Entry, Office; MF-3 (bedroom hall, west wall north of the fireplace, 8 loops) serves Living, Bedroom 1, Bath 1, Closet (Bedroom 2), Bedroom 2; MF-4 (bedroom hall, west wall south of the fireplace, 6 loops) serves Living, Bath 2, Bedroom 3. HS-1 in the mechanical closet feeds all seven on six 1" insulated pairs under the boards, 192' each way in all; MF-2 and MF-G share one pair rising in the garage wall.

6. The leaders rise straight up out of the slab at their port positions into the cabinet (sleeve each rise in 12" of pipe insulation where it leaves the concrete) and bundle up to the manifold; they may be gathered sideways in the last 12" below the cabinet. Cabinets are recessed in the kitchen and entry walls (MF-K, MF-5) and the hall wall (MF-3, MF-4) and surface-mounted on the laundry's sound wall (MF-1) and both faces of the garage wall (MF-2, MF-G).
7. Zoning is by thermostat: four zones (master suite, great room and utility, bedroom wing and office, garage) on 24 V actuators at each loop's manifold port, so a room can get its own thermostat later by rewiring, without new tube.
8. Flow: 11.9 gpm with every loop calling, each loop set on its flow meter at a 10°F drop. Design heat 52,119 Btu/h, of which 10,539 is the garage at 50°F; 7% is added for heat lost down through the R-18 boards and out the slab edge. Rooms with no field of their own are heated by the leader bundles running through them: Powder (16' of leaders for 150 Btu/h), Bedroom Hall (141' of leaders for 422 Btu/h), Bedroom 1 Closet (1' of leaders for 86 Btu/h). Nothing heats the Mechanical (493 Btu/h), Coat Closet (30 Btu/h), Closet (Bedroom 3) (916 Btu/h); they borrow from the rooms around them.
9. Water: 110°F supply at -11°F outside, reset lower as it warms, so a condensing boiler stays condensing. Floor surfaces stay at or under 110°F. At that supply L31 (Closet (Bedroom 2), Bath 1) falls about 780 Btu/h short at design; see the open items.
10. Pump: the hardest loop is L11 at 1.9 ft in the tube; add about 3 ft for the feed pair, manifold and actuator. A constant-ΔP ECM circulator at

about 5 ft covers the house; the garage has its own.

11. Under the boards: 10-mil vapor barrier over the compacted base, lapped and taped; R-10 (2") rigid foam at the slab edge. The 4" slab goes over the pucks with the tube tops about 1" below the surface of the boards; no mesh ties are needed, the pucks hold the tube. Pressure test every loop at 100 psi water or 60 psi air for 24 hours before the pour and keep it under pressure through the pour. Photograph every room with a tape in frame before concrete.
12. HS-1 is circuit 27 on E-1 (200 A panel in the garage); E-1 also gives each manifold cabinet an outlet for its actuators and the zone controller.

Loop schedule

Ports are the supply and return positions counted from the low end of the manifold's row. Leader is each way. Heat includes the share lost downward. Pick a row to find the loop.

Loop	Zone	Rooms	Manifold	Ports	In field	Leader	Total	Btu/h	Flow	Head	Floor
L1	1	Master Bed	MF-1	14/13	152'	12'	177'	1,191	0.24 gpm	0.6 ft	76°F
L2	1	Master Bed	MF-1	12/11	121'	8'	138'	922	0.20 gpm	0.3 ft	76°F
L3	1	Master Bed	MF-1	10/9	121'	5'	131'	878	0.20 gpm	0.3 ft	76°F
L4	1	Master Bath	MF-1	5/6	61'	17'	96'	1,930	0.39 gpm	0.7 ft	91°F
L5	1	Master Bath	MF-1	2/1	75'	9'	93'	1,910	0.38 gpm	0.7 ft	91°F
L6	1	Master Closet	MF-1	3/4	128'	15'	159'	1,767	0.35 gpm	1.0 ft	82°F
L7	2	Laundry	MF-1	7/8	36'	4'	45'	505	0.20 gpm	0.1 ft	82°F
L8	1	Master Hall	MF-2	5/6	36'	27'	92'	522	0.20 gpm	0.2 ft	79°F
L9	2	Mud Room	MF-2	1/2	59'	3'	67'	440	0.20 gpm	0.2 ft	77°F
L10	2	Mud Room	MF-2	7/8	50'	17'	85'	566	0.20 gpm	0.2 ft	77°F
L11	2	Dining	MF-K	2/3	112'	35'	182'	2,356	0.47 gpm	1.9 ft	85°F
L12	2	Dining	MF-K	4/5	112'	30'	172'	2,186	0.44 gpm	1.6 ft	85°F
L13	2	Kitchen	MF-K	6/7	128'	25'	179'	986	0.20 gpm	0.4 ft	75°F
L14	2	Kitchen	MF-K	8/9	54'	18'	91'	490	0.20 gpm	0.2 ft	75°F
L15	2	Kitchen	MF-K	10/11	77'	10'	97'	527	0.20 gpm	0.2 ft	75°F
L16	2	Pantry	MF-5	1/2	98'	3'	105'	1,377	0.28 gpm	0.4 ft	83°F
L17	2	Entry	MF-5	11/12	22'	6'	36'	432	0.20 gpm	0.1 ft	83°F
L18	2	Living	MF-5	3/4	148'	9'	166'	1,185	0.24 gpm	0.5 ft	83°F

Loop	Zone	Rooms	Manifold	Ports	In field	Leader	Total	Btu/h	Flow	Head	Floor
L19	2	Living	MF-5	5/6	148'	12'	173'	1,246	0.25 gpm	0.6 ft	83°F
L20	2	Living	MF-5	7/8	148'	16'	180'	1,328	0.27 gpm	0.7 ft	83°F
L21	2	Living	MF-3	6/5	94'	17'	128'	787	0.20 gpm	0.3 ft	76°F
L22	2	Living	MF-3	4/3	107'	20'	147'	934	0.20 gpm	0.3 ft	76°F
L23	2	Living	MF-4	1/2	66'	7'	82'	522	0.20 gpm	0.2 ft	76°F
L24	2	Living	MF-4	3/4	66'	10'	88'	559	0.20 gpm	0.2 ft	76°F
L25	2	Living	MF-3	1/2	30'	11'	53'	311	0.20 gpm	0.1 ft	76°F
L26	2	Living	MF-4	5/6	26'	6'	38'	223	0.20 gpm	0.1 ft	76°F
L27	3	Office	MF-5	13/14	116'	17'	151'	1,409	0.28 gpm	0.7 ft	83°F
L28	3	Office	MF-5	15/16	93'	12'	119'	1,110	0.22 gpm	0.3 ft	83°F
L29	3	Bedroom 1	MF-3	7/8	134'	14'	164'	1,857	0.37 gpm	1.1 ft	82°F
L30	3	Bedroom 1	MF-3	9/10	134'	18'	171'	2,018	0.40 gpm	1.4 ft	82°F
L31	3	Closet (Bedroom 2), Bath 1	MF-3	13/14	63'	32'	128'	2,455	0.49 gpm	1.5 ft	110°F
L32	3	Bedroom 2	MF-3	15/16	146'	22'	190'	1,291	0.26 gpm	0.7 ft	77°F
L33	3	Bedroom 2	MF-3	17/18	122'	16'	154'	1,029	0.21 gpm	0.4 ft	77°F
L34	3	Bath 2	MF-4	7/8	42'	7'	56'	895	0.20 gpm	0.1 ft	88°F
L35	3	Bedroom 3	MF-4	14/13	128'	9'	146'	1,654	0.33 gpm	0.8 ft	81°F
L36	3	Bedroom 3	MF-4	12/11	128'	15'	158'	1,782	0.36 gpm	1.0 ft	81°F
L37	G	Garage	MF-G	1/2	110'	19'	149'	1,038	0.21 gpm	0.4 ft	57°F
L38	G	Garage	MF-G	3/4	110'	18'	147'	1,021	0.20 gpm	0.4 ft	57°F
L39	G	Garage	MF-G	5/6	54'	16'	88'	598	0.20 gpm	0.2 ft	57°F
L40	G	Garage	MF-G	7/8	142'	19'	181'	1,264	0.25 gpm	0.6 ft	57°F
L41	G	Garage	MF-G	9/10	94'	17'	128'	888	0.20 gpm	0.3 ft	57°F
L42	G	Garage	MF-G	11/12	94'	15'	125'	867	0.20 gpm	0.3 ft	57°F
L43	G	Garage	MF-G	13/14	142'	11'	165'	1,149	0.23 gpm	0.5 ft	57°F
L44	G	Garage	MF-G	15/16	94'	9'	112'	773	0.20 gpm	0.3 ft	57°F
L45	G	Garage	MF-G	23/24	120'	4'	129'	892	0.20 gpm	0.3 ft	57°F
L46	G	Garage	MF-G	34/33	113'	10'	135'	935	0.20 gpm	0.3 ft	57°F
L47	G	Garage	MF-G	32/31	113'	23'	160'	1,114	0.22 gpm	0.5 ft	57°F
Zone 1	1	Master suite	MF-1, MF-2		7 loops		887'	9,119	2.0 gpm		
Zone 2	2	Great room and utility	MF-1, MF-2, MF-3, MF-4, MF-5, MF-K		19 loops		2,114'	16,960	4.5 gpm		
Zone 3	3	Bedroom wing and office	MF-3, MF-4, MF-5		10 loops		1,438'	15,501	3.1 gpm		

Loop	Zone	Rooms	Manifold	Ports	In field	Leader	Total	Btu/h	Flow	Head	Floor
Zone G	G	Garage (held at 50°F)	MF-G		11 loops		1,518'	10,539	2.3 gpm		

Manifolds

Each station's row of 3" port positions along its wall; a loop takes two next to each other.

Tag	Loops	Ports used	Where	Zones	Feed
MF-1	7 loops: L1, L2, L3, L4, L5, L6, L7	14 of 14	Laundry, on the north wall west of the dryer	1, 2	1" pair F-1, 41' each way
MF-2	3 loops: L8, L9, L10	6 of 8	Mud room, on the south wall (garage wall), west end	1, 2	1" pair F-2, 27' each way
MF-G	11 loops: L37, L38, L39, L40, L41, L42, L43, L44, L45, L46, L47	22 of 34	Garage, on the mud room wall (two 7-loop manifolds side by side)	G	1" pair F-2, 27' each way (shared with MF-2)
MF-K	5 loops: L11, L12, L13, L14, L15	10 of 12	Mud room, east wall (kitchen wall), south of the powder room	2	1" pair F-K, 17' each way
MF-5	7 loops: L16, L17, L18, L19, L20, L27, L28	14 of 17	Pantry, on the east wall (entry wall)	2, 3	1" pair F-5, 12' each way
MF-3	8 loops: L21, L22, L25, L29, L30, L31, L32, L33	16 of 18	Bedroom hall, west wall north of the fireplace	2, 3	1" pair F-3, 52' each way
MF-4	6 loops: L23, L24, L26, L34, L35, L36	12 of 14	Bedroom hall, west wall south of the fireplace	2, 3	1" pair F-4, 43' each way

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The installer takes off the final list.

Item	Qty	Notes
1/2" O2-barrier PEX, 1,000' coils	7 coils	5,957' in 47 uncut lengths of 36' to 190' (the schedule), cut longest first
EPS25 Super 4 nub board, 4' x 4'	274 boards (55 bundles of 5)	4,167 sq ft of slab + 5% for cuts; R-18, 1/2" tube channels on 3" centres
1" PEX supply + return, insulated	404 lf	HS-1 to the stations, in the base course under the boards
Stainless manifolds with flow meters	7 stations	MF-1 7-loop, MF-2 3-loop, MF-G 11-loop, MF-K 5-loop, MF-5 7-loop, MF-3 8-loop, MF-4 6-loop (MF-G as two manifolds side by side)
24 V thermal actuators	47	one per loop, on the return port
Zone controller + thermostats	1 + 4	T1, T2, T3, TG; outdoor reset sensor
Circulators	2	P-R1 constant ΔP for the house, P-RG for the garage with its mixing valve
Manifold cabinets	7	recessed at MF-K, MF-5, MF-3 and MF-4; surface at MF-1, MF-2 and MF-G
R-10 rigid foam, 2", slab edge	346 lf	the perimeter of the slab, full depth
10-mil vapor barrier	4,584 sf	under the boards, lapped and taped
Pipe insulation, 1/2" x 12"	94 pcs	one sleeve on every tube end where it rises out of the concrete